Potential Risk of Lead Exposure May Indicate Poverty Levels*

Megan Joseph

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Many Americans live below the poverty line. As such, one might be interested in what impacts poverty. Using data from the Calenviroscreen 4.0 tool, I investigate how the potential risk of lead exposure to children in low-income homes may impact the poverty levels in a California census tract. I fit a simple linear regression model with the percent of the population living with an income less than $87,000 for two adults and two children as the response and the potential risk of lead exposure to children in low income homes as the predictor. A hypothesis test showed that there would be a linear association if the conditions were met, but it may be more impactful to do multiple regression because of the nuances surrounding poverty.

1 Introduction

In 2023, around 6.4 million people in California were living below the California Poverty Measure poverty line ($43,990 annually for two adults and two children) (Bohn et al. 2025). Identifying what may impact poverty may bring us closer to ending it. One potential impact is lead exposure, although poverty is a very complex issue and is a culmination of a variety of factors. Children are more susceptible to the health effects that come from lead exposure due to how they are exposed, their developing brains, and how the lead is absorbed (August 2021). Many California houses were built before the lead-based paint ban in 1978 and still remains a major source of lead exposure to young children (August 2021). Additionally, around 25% of children in California live in poverty. It has already been shown that older housing, which often has lead-based paint, and poverty are associated with high blood lead levels (August 2021). Based on this, I aim to answer the research question: is there a linear association between the potential risk for lead exposure in children in low-income communities and the percent of population living below two times the federal poverty level.

*Project repository available at: https://github.com/meganajoseph/math261a_paper1.

As a student pursuing their Masters in Statistics, there are many things I still don't know. I am not very well-versed in the social sciences which may impact the results of this study. Additionally, I am still learning about linear regression as a whole which may also impact the study.

In this study, I use data from CalEnviroScreen 4.0 in an attempt to answer the research question. CalEnviroScreen is a model that was designed to provide information on the state of California and is made up of the components Pollution Burden and Population Characteristics (Kutner et al. 2005). The model assigns a score to each geographic location based on the components (Kutner et al. 2005). This tool was created in order to assess which communities are more impacted by the impacts of pollution and other socioeconomic stressors in order to better support them (Kutner et al. 2005).

I used the Poverty and Lead columns to fit a simple linear regression model with Poverty as the response and Lead as the predictor. Lastly, I perform a hypothesis test to determine if there really is a linear association between the two.

The results of the hypothesis test indicate that there is a linear association. This is important because it suggests that the potential risk for lead exposure can be used to measure poverty. Despite this, a multiple linear regression model may be a better fit for this study.

The remainder of the paper is structured as follows: Section 2 discusses the data, Section 3 discusses the model, and Section 4 presents the results and their implications.

2 Data

2.1 Data Overview

The data is from CalEnviroScreen 4.0 which is a tool created by the Office of Environmental Health Hazard Assessment in the California Environmental Protection Agency used to help identify communities in California that are most affected by pollution (August 2021). The data used for the tool come from national and state sources and was last updated in 2021 (Wieland 2021).

Each of the 8035 rows represents a census tract from 2010. A census tract is made up of the smallest geographic unit for population data: census blocks (August 2021). They are typically areas bounded by streets, streams, and other features (Rossiter 2021). The key variables are Lead, which quantifies the potential risk for lead exposure in children in low-income communities, and poverty, which represents the percent of the population living below two times the federal poverty level (August 2021). Lead is transformed to a numeric variable because it is originally a character. Additionally, Lead ranges from 0 - 100 due to how it was created. See Section 2.3 for more details. Many other variables are present in the dataset but will not be used for this paper. Figure 1 shows a scatter plot of Poverty on the y-axis and

Lead on the x-axis. As the risk for lead exposure increases, the percent in poverty increases a bit as well.

Since this data is at the census tract level rather than at the household level, this analysis cannot conclude anything on the household scale, only at the census tract level. This analysis is not equivalent to an analysis at an individual household level. While there may be similar trends across both, the conclusions will differ. This is because we lose some variation at the household level when data is aggregated in this way. Since data is only available at the census tract level, the analysis will proceed at that scale knowing that analysis cannot be made on the household level.

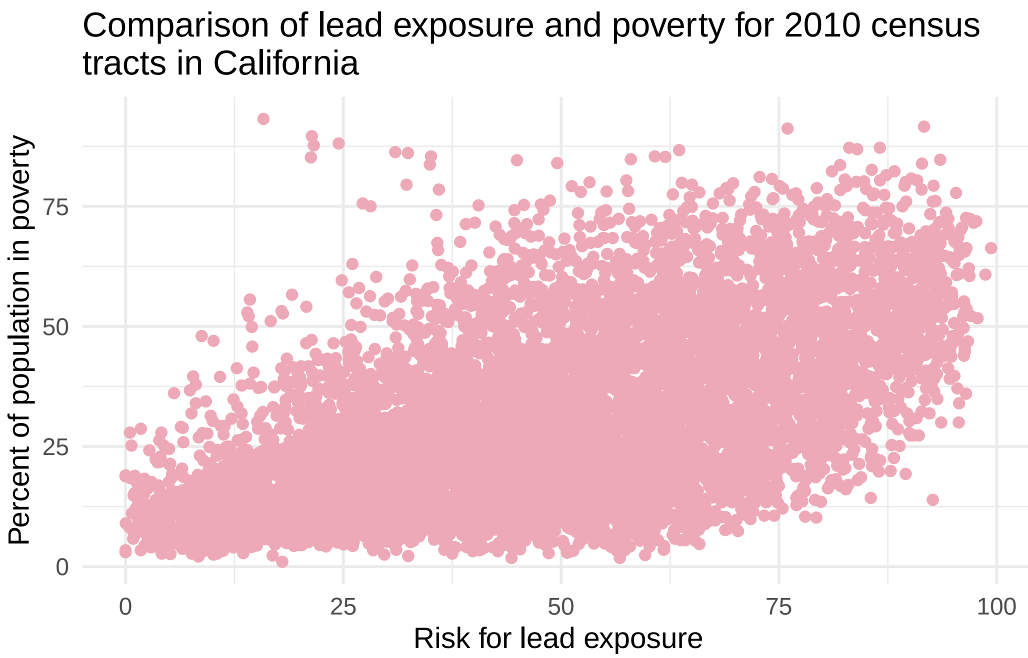

Figure 1: Scatter plot of the risk of lead exposure (x-axis) and percent of population in poverty (y-axis).

2.2 Poverty

The poverty column was taken from the American Community Survey (ACS), an ongoing survey of a sample of the US population by the US Census Bureau, from 2015-2019 (August 2021). While the survey uses the federal poverty level to determine if people live below the federal poverty line, this dataset uses twice the federal poverty line because of California's high cost of living (August 2021). As a result, Poverty is calculated by dividing the total population with a poverty status with the number of individuals below 200% of the poverty level for each census tract in California rather than 100% (August 2021).

One issue with the dataset is that the ACS data come from a sample of the population rather than the entire population, so the Poverty variable may not be fully representative of the California population. To increase reliability, census tracts with estimates that either had a relative standard error less than 50 or a standard error less than the mean standard error of all California census tract estimates for poverty were included and the rest received no score (August 2021).

2.3 Lead

Another potential issue is that there isn't direct information on exposures of lead, so the values used are based on pollution sources, releases, and environmental concentrations for potential exposures (August 2021). To calculate this value, they used the percentage of households in each census tract with a likelihood of lead-based paint hazards based on housing age and the percentage of low-income households with children (August 2021). The percentage of homes with a likelihood of lead-based paint hazards was calculated in a series of steps. First, California homes were grouped into categories based on age (August 2021). Then, the number of housing units in each category was multiplied by the reported percentage of homes with lead-based paint hazards (August 2021). Finally, the number of housing units in each category were added and divided by the total number of housing units in the census tract (August 2021). The percentage of low income households with children was calculated by estimating the number of households with incomes less than 80% of the county mean with one or more children under the age of six (August 2021). The final lead risk for housing score is the weighted sum of the percent of homes with likelihood of lead-based paint hazards percentile and the low-income household with children percentile: $0.6L + 0.4H = S$ where L is the percentile for lead-based paint hazards, H is the percentile of low-income households with children, and S is the final score (August 2021). This creates a lowest value of 0 when both L and H are 0 and a highest value of 100 when both L and H are 100. I will assume that the indicators created are indicative of the potential for lead exposure.

2.4 Alternative Sources

Alternative sources of data include finding data directly from the US Census so as to use data from the entire population rather than a sample. Additionally, finding data on the household level would allow conclusions per household rather than per census tract. Using a dataset that had actual measures of lead exposure in housing would also lead to more reliable answers.

3 Methods

To answer the research question, I will fit a simple linear regression model to the data with Poverty as the response and Lead as the predictor. Let Y_i represent the percent of the population in the census tract living below two times the federal poverty level and X_i represent the potential risk for lead exposure in children living in low-income communities with older housing. The model can we written as

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \text{ for i = 1, ..., n}$$

where β_0 represents the mean of the probability distribution of the percent of the population in poverty when risk for lead exposure is 0, β_1 represents the average change in percent in poverty per unit increase in risk for lead exposure, and ε_i represents independent random error assumed to have mean 0 and variance σ^2.

The assumptions for linear regression are:

1. The variables used accurately reflect the quantities of interest, the model should include all predictors, and the model should be applicable for the research question.
2. The sample data is representative of the population of interest.
3. The mean function must be correct.
4. Errors must be independent.
5. The variance of the errors must be equal.
6. Errors must be normal.

3.1 Addressing Assumptions

1. The variables used here are Poverty and Lead from the dataset. They reflect the quantities of interest: the percent living under two times the federal poverty line and the potential risk of lead exposure. The model includes Lead as a predictor.

2. The population of interest is a California census tract. The sample data comes from each census tract in California which is representative of our population of interest. Certain areas, though, may not be accounted for in the Lead column due to ACS taking data from only a sample of the US population.

3. There is a slight departure from linearity as seen in Figure 2 (a) where the residuals are not equally dispersed around the line $y = 0$. This means that both the parameter estimates and error variance will be biased (Kutner et al. 2005). A curvilinear line may be better suited, but I will still fit a simple linear regression.

4. Knowing information about one census tract will tell us information on the surrounding tracts because they do not contain huge geographic areas. Since their demographics are similar, their scores will be similar as well. As a result, the errors are not independent. This means that the estimators are unbiased but have biased variance (Kutner et al. 2005).

5. Figure 2 (a) shows that the variance of the errors may not be equal. It shows a slight cone shape, especially below the $y = 0$ line. Nonconstancy of error variance is very common, but it does lead to less efficient estimates and invalid error variance estimates (Kutner et al. 2005).
6. A Q-Q plot of the residual percentiles against the standard normal percentiles is the best way to check for normality because we can see direct departures from the normal distribution. Figure 2 (b) shows that the errors appear to be close to normal with upward deviations on each end. I will assume that the errors are normal.

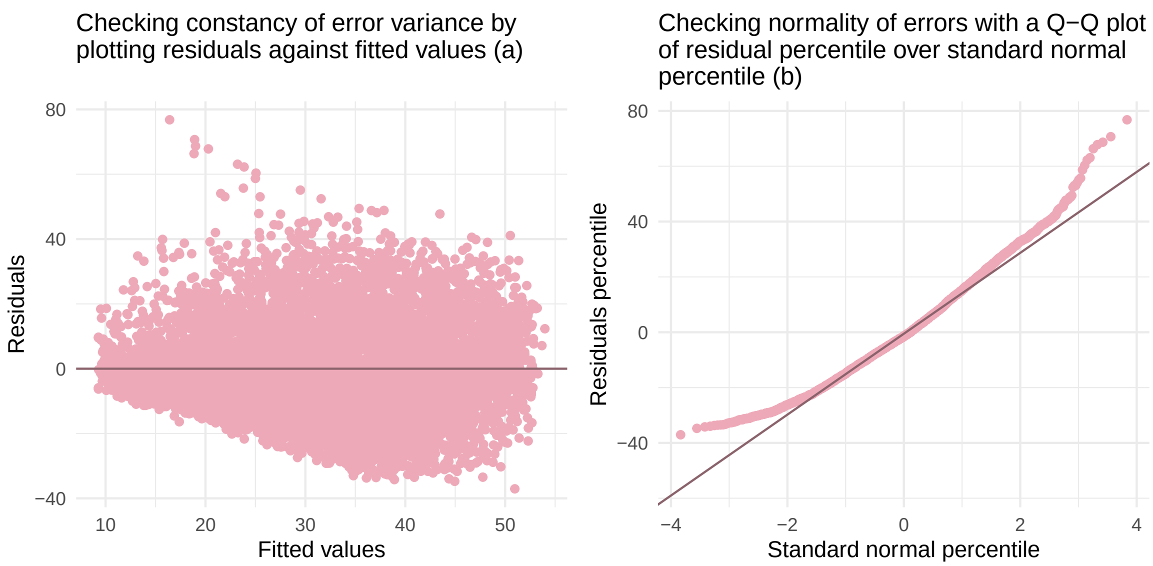

Figure 2: Plot a shows unequal variance in error terms and plot b shows deviations from normality on both ends of the residual percentile.

I use the `lm()` function from the R programming language (R Core Team 2025) to fit the linear regression model.

4 Results

The estimated slope parameter is $b_1 \approx 0.45$. This means that for each unit increase in potential risk for lead exposure, the percent in poverty increases by 0.45 on average. The estimated intercept parameter is $b_0 \approx 9.287$. Since the dataset includes a potential risk for lead exposure at 0, the average percent in poverty at that value is 9.287.

The coefficient of determination (R^2) is 0.326. This means that 32.6% of the variation in percent of households living two times below the federal poverty level is explained by the potential risk of lead exposure. This is not very high which is most likely due to poverty being a much more complex issue that cannot be explained by potential lead exposure alone. Other possible factors of poverty include, but are not limited to, race, education, neighborhood, and parent income levels. Figure 3 plots the regression line on top of the data. Many of the points are far above and far below the line which only heightens the fact that potential risk for lead exposure is not the only explanatory variable for poverty.

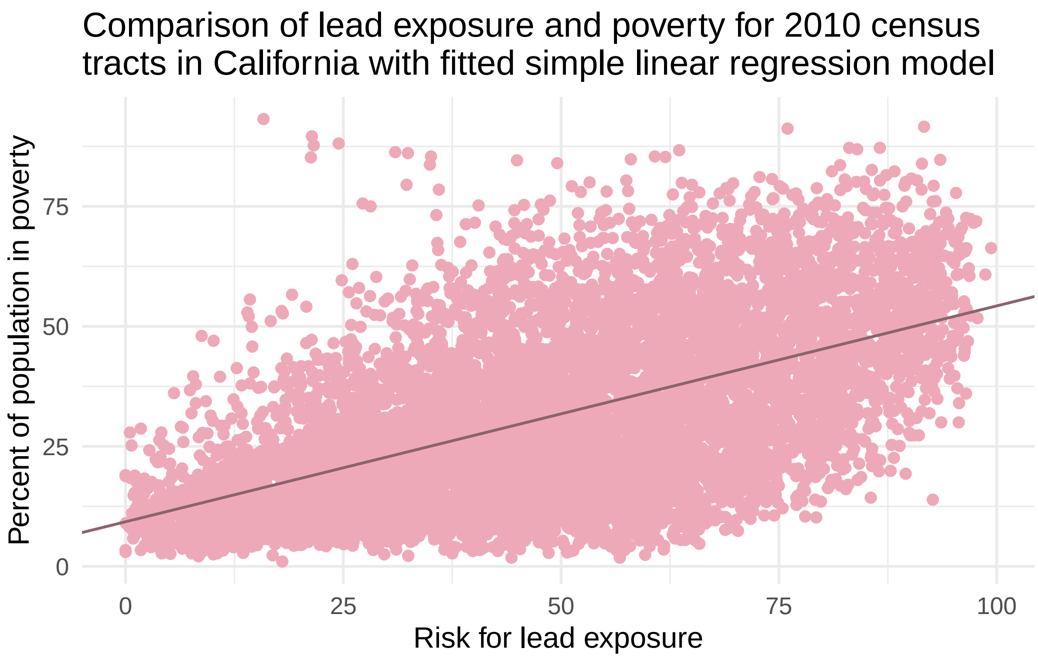

Figure 3: Scatter plot of the risk of lead exposure (x-axis) and percent of population in poverty (y-axis) with fitted simple linear regression model $Y_i = b_0 + b_0 X_i$.

4.1 Hypothesis Test

A two-sided hypothesis test of the form

$$H_0 : \beta_1 = 0$$
$$H_a : \beta_1 \neq 0$$

will be used to test if there is a linear association between Poverty and Lead. If we conclude the null hypothesis, then there is no linear association (Kutner et al. 2005).

To do this, we assume the normal error regression model (Kutner et al. 2005). Earlier, I found that the errors are approximately normal, but the variance does not appear to be constant, it is possibly non-linear, and the errors are not independent. These caveats decrease the trustworthiness of the hypothesis test. However, for the purposes of this analysis, I will proceed as if the necessary assumptions were met.

I will control the risk of a Type I error at $\alpha = 0.05$. With this, I will make a 95% confidence interval. If 0 is contained in the interval, then I will conclude H_0. Otherwise, I will conclude H_a (Kutner et al. 2005).

The 95% confidence limits for β_1 are $b_1 \pm t(1-\alpha/2; n-2)s\{b_1\}$ where $t(1-\alpha/2; n-2)$ is the $\alpha/2$ 100 percentile of the t distribution with $n-2$ degrees of freedom and $s\{b_1\}$ is the point estimator of the standard deviation (Kutner et al. 2005). Plugging in our numbers, the 95% confidence limits for β_1 are 0.45 $\pm t$(0.975; 7934) $\cdot$ 0.007

After calculating, $0.436 \leq \beta_1 \leq 0.464$. Since zero is not contained in our interval, we can reject the null hypothesis. Therefore, if the necessary assumptions were met, there would be a linear association between Poverty and Lead.

4.2 Discussion

The aim of this paper was to answer the following research question: Is there a linear association between potential risk of lead exposure and poverty levels? I investigated this with data from CalEnviroScreen 4.0. Then, I created a simple linear regression model using Lead as the predictor and Poverty as the response. The final model is $Y = 9.287 + 0.45 \cdot X$. The model explained 32.6% of the variation in percent of households living two times below the federal poverty level is explained by the potential risk of lead exposure. Lastly, I performed a hypothesis test and created a 95% confidence interval in order to determine if there would be a linear association under the correct assumptions. The resulting confidence interval is $0.436 \leq \beta_1 \leq 0.464$ which does not include zero showing that there would be a linear association.

Since my findings show a linear association, the potential risk for lead exposure in children in low-income communities leads to some indication of poverty, though not all of it.

A potential weakness of this study is that Lead may be correlated with Poverty since it is based on low-income housing. Additionally, an exponential model may be a better fit than a linear one based on the scatter plot, which also shows high variance. Since the data is at the census tract level, the results can only be interpreted at the same level and no conclusions can be made at the individual household level.

Potential extensions include creating a transformed multiple regression model with a few predictors, possibly socioeconomic ones as well. Using data for lead exposure that does not take into account low-income communities is a potential improvement. This study can also be extended to the entire United States population rather than just California.

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